

**F TALENT CODE 2026****Automated Drone Patrol Station Assignment***Time Limit: 1 second(s) | Memory Limit: 512 MB***Problem Statement**

In a high-tech smart city grid of size  $k \times k$ , a security company wants to deploy two surveillance drones to guard critical sectors. To prevent signal interference and physical collisions, no two drones can be stationed such that they cover each other's offset patrol routes. Specifically, two drones interfere with each other if their grid positions form an L-shape jump (moving 2 units in one axis and 1 unit in the other axis). Your task is to compute, for each grid size  $k$  from 1 to  $n$ , the total number of ways to place the two drones safely on the  $k \times k$  grid without signal interference.

**Input**

A single integer  $n$  ( $1 \leq n \leq 10^4$ ).

**Output**

Print  $n$  integers, where the  $k$ -th integer represents the number of valid drone placements for  $k \times k$  grid.

**Constraint(s)**

$1 \leq n \leq 10^4$ .

**Example 1:**

| Input | Output       |
|-------|--------------|
| 3     | 0<br>6<br>28 |

**Explanation:**

$k = 1$ : A  $1 \times 1$  board has only 1 square, which is not enough space to place 2 drones  $\rightarrow$  0 ways.

$k = 2$ : A  $2 \times 2$  board has 4 squares. Choosing any 2 out of the 4 squares gives  $C(4,2) = 6$  ways. Since  $2 \times 2$  board is too small to form an L-shape ( $2 \times 3$  or  $3 \times 2$ ), no positions cause signal interference  $\rightarrow$  Result = 6 ways.

$k = 3$ : A  $3 \times 3$  board has 9 squares. The total number of ways to choose 2 squares is  $C(9,2) = 36$  ways. The number of L-shapes ( $2 \times 3$  or  $3 \times 2$ ) that can be formed on a  $3 \times 3$  board is 4 rectangles, with each rectangle containing 2 pairs of interfering squares  $\rightarrow 4 * 2 = 8$  interfering pairs. Result =  $36 - 8 = 28$  ways.

| Example 2: |                              |
|------------|------------------------------|
| Input<br>4 | Output<br>0<br>6<br>28<br>96 |

**Explanation:**

With cases  $k = 1, 2, 3 \Rightarrow$  the same as Example 1.

With  $k = 4$ : A  $4 \times 4$  grid has 16 squares. The total number of distinct pairs of squares is  $C(16, 2) = 120$  ways. The number of  $2 \times 3$  rectangles is  $(4-1)(4-2) = 6$ . The number of  $3 \times 2$  rectangles is  $(4-2)(4-1) = 6$ . The total number of interfering pairs is  $(6 + 6) * 2 = 24$  pairs. Result =  $120 - 24 = 96$  ways.